

OFFICIAL ARCHITECTURE & IMPLEMENTATION AUDIT

NEXTRON26 SYMPOSIUM WEBSITE

Complete Current Implementation Report

Institution: Kuppam Engineering College (KEC) • **Department:** Electronics and Communication Engineering (ECE) • **Status:** PRODUCTION-READY • **Audited:** March 2026

1. PROJECT OVERVIEW

NEXTRON-2026 is the national-level technical symposium web portal for the Department of ECE at Kuppam Engineering College. The platform manages technical and non-technical event registrations, dynamic fee calculations, peer-to-peer UPI payments with transaction validation, college ID & payment proof uploads, multi-tiered administrator & coordinator portals, and automated real-time Google Sheets synchronization.

- **Frontend Stack:** React 19, TypeScript 5.7, Vite 8, Tailwind CSS v4, Lucide React, SheetJS (XLSX).
- **Backend Stack:** Supabase PostgreSQL 17 (ap-northeast-2), Supabase Edge Functions (Deno runtime).
- **Authentication:** Custom PostgreSQL session engine with bcrypt (pgcrypto) hashing & 24h bearer tokens.
- **Storage:** Two private Supabase Storage buckets (id-cards, payment-proofs) with 50 KB strict upload limits.
- **Hosting:** Vercel Production Deployment (<https://symposium-website-steel.vercel.app>).

2. SYSTEM ARCHITECTURE & CREDENTIAL DIRECTORY

The system operates a dual-role access control model. Below are the verified active login user identifiers:

ADMINISTRATOR

User ID: admin@nextron.com (Access: Global across all 17 events & coordinator management)

Coordinator #	Assigned Event	User ID (Login Email)	Access Scope
Coordinator 1	Paper Presentation	paper@nextron.com	Scoped to Paper Presentation
Coordinator 2	Project Expo	project@nextron.com	Scoped to Project Expo
Coordinator 3	Code Debugging	codedebug@nextron.com	Scoped to Code Debugging
Coordinator 4	Tech Quiz	techquiz@nextron.com	Scoped to Tech Quiz
Coordinator 5	Logo Design	logodesign@nextron.com	Scoped to Logo Design
Coordinator 6	Ideathon	ideathon@nextron.com	Scoped to Ideathon
Coordinator 7	Web Design	webdesign@nextron.com	Scoped to Web Design
Coordinator 8	Electro Charades	electrocharades@nextron.com	Scoped to Electro Charades
Coordinator 9	AI Video Animation & Generation	aivideo@nextron.com	Scoped to AI Video Animation
Coordinator 10	Free Fire	freefire@nextron.com	Scoped to Free Fire
Coordinator 11	BGMI	bgmi@nextron.com	Scoped to BGMI
Coordinator 12	Photography	photo@nextron.com	Scoped to Photography
Coordinator 13	Treasure Hunt	treasure@nextron.com	Scoped to Treasure Hunt
Coordinator 14	Reels Making	reels@nextron.com	Scoped to Reels Making
Coordinator 15	Meme Making	meme@nextron.com	Scoped to Meme Making
Coordinator 16	Cine Quiz	cinequiz@nextron.com	Scoped to Cine Quiz
Coordinator 17	Act & Guess	actguess@nextron.com	Scoped to Act & Guess

3. WEBSITE PAGES & ROUTING

Page / Route	Access	Functionality	Status
Home (/)	Public	Symposium introduction, countdown timer, leadership patrons, faculty directory	WORKING
Events (/)	Public	Technical/Non-Technical event cards, rules modal, registration form with dynamic total	WORKING
Registration Status (/)	Public	Lookup by NEX-XXXXXX registration code; live slot booking & verification status	WORKING
Login (/login)	Public	Admin & Coordinator dual login portal; authenticates via app_login RPC	WORKING
Admin Dashboard (/admin)	Role: Admin	Overview across all 17 events, revenue statistics, search, Excel export, coordinator CRUD	WORKING
Coordinator Dashboard (/coordinator)	Role: Coordinator	Scoped to assigned events; reviews UTR, views ID & proof, ACCEPT / REJECT actions	WORKING
Document Viewer (/view-id)	Authenticated	Generates secure short-lived signed URLs (1h) for private storage objects	WORKING

4. EVENTS DIRECTORY & PAYMENT CONFIGURATION

Event Name	Category	Fee Model	Team Size	UPI VPA	QR Asset
Paper Presentation	Technical	₹50 / head	1–2 members	raghavij2006@okaxis	/qr/paper.png
Project Expo	Technical	₹100 / head	3–4 members	aswinappu2005@yb1	/qr/project.png
Code Debugging	Technical	₹50 / head	1–2 members	9985708284-2@yb1	/qr/codedebug.png
Tech Quiz	Technical	₹50 / head	3–4 members	9100309531-2@yb1	/qr/techquiz.png
Logo Design	Technical	₹50 / head	1–2 members	7842406485@ax1	/qr/logodesign.png
Ideathon	Technical	₹50 / head	2–3 members	udaykiran9392@ax1	/qr/ideathon.png
Web Design	Technical	₹50 / head	1–2 members	7670973554-2@yb1	/qr/webdesign.png
Electro Charades	Technical	₹50 / head	3–4 members	sowjanya0583@yb1	/qr/electrocharades.png
AI Video Animation & Generation	Technical	₹50 / head	1–2 members	6281325750@superyes	/qr/aivideo.png
Free Fire	Non-Technical	₹200 / team	4 members	9959163510@ib1	/qr/freefire.png
BGMI	Non-Technical	₹200 / team	4 members	9347630150-3@ib1	/qr/bgmi.png
Photography	Non-Technical	₹100 / head	Individual	7995985503@yb1	/qr/photography.png
Treasure Hunt	Non-Technical	₹50 / head	3–4 members	6301861219-3@ib1	/qr/treasure.png
Reels Making	Non-Technical	₹50 / head	3–4 members	6304958303@yb1	/qr/reels.png
Meme Making	Non-Technical	₹50 / head	2–3 members	raavishyamsriram@yb1	/qr/meme.png
Cine Quiz	Non-Technical	₹50 / head	3–4 members	9908382478-2@yb1	/qr/cinequiz.png
Act & Guess	Non-Technical	₹50 / head	3–4 members	9390596679@ptsbi	/qr/actguess.png

5. REGISTRATION & PAYMENT VERIFICATION FLOW

STUDENT SUBMISSION

Select Event & Agree to Rules → Enter Team & Member Details → Upload ID Cards (≤ 50 KB)
 → Scan Event UPI QR & Copy VPA → Complete UPI Payment → Enter UTR → Upload Payment Proof (≤ 50 KB)
 → Submit Form → Database creates Registration, Member, and Payment rows (Status: PENDING)
 → Receive NEX-XXXXXX & WhatsApp Group Link

COORDINATOR / ADMIN VERIFICATION

Login → Dashboard displays scoped registrations → Inspect UTR, ID card & payment proof screenshot
 • Click ACCEPT: Sets registration_status = 'ACCEPTED', payment_status = 'VERIFIED'
 • Click REJECT: Sets registration_status = 'REJECTED', payment_status = 'FAILED'
 → Trigger invokes sync-google-sheet Edge Function & updates Google Sheets in background

6. DATABASE SCHEMA & STORAGE SPECIFICATIONS

- **events:** id (PK), name, fee_per_head, fee_type (per_head / per_team), min_members, max_members, is_active.
- **registrations:** id (UUID PK), registration_number (NEX-XXXXXX), event_id (FK), team_name, college_name, amount, registration_status (PENDING/ACCEPTED/REJECTED).
- **members:** id (UUID PK), registration_id (FK), member_number, full_name, phone, email, id_card_path.
- **payments:** id (UUID PK), registration_id (FK), amount, utr_number, payment_status (PENDING/VERIFIED/REJECTED/FAILED), payment_proof_path, remarks, verified_at, verified_by.
- **coordinators:** id (UUID PK), name, email (UNIQUE), password_hash, role (admin/coordinator), is_active, assigned_events (TEXT[]).
- **app_sessions:** id (UUID PK), user_id (FK), role, token (32-byte hex), expires_at (24 hours).
- **Storage Buckets:**
 - id-cards: Private, 50 KB limit, MIME: image/jpeg, image/png.
 - payment-proofs: Private, 50 KB limit, MIME: image/jpeg, image/png, image/webp.

7. GOOGLE SHEETS SYNCHRONIZATION

Synchronized in real-time via PostgreSQL triggers (pg_net extension) calling the sync-google-sheet Edge Function with service account JWT authentication:

- **Tabs Configured:** 17 event-specific tabs + 1 Payment Summary tab.
- **Row Data Recorded:** Registration Number, Event, Team, College, Team Size, Amount Paid, Payment Status, Registration Status, Payment Method, Collection Date/Time, UTR, Clickable Payment Proof Link, Clickable Member ID Card Links, and Member contact details.
- **Deduplication:** In-place row updates on Registration Number match; appends row if new.

8. AUDIT STATUS & SUMMARY

OVERALL STATUS: WORKING / PRODUCTION-READY

- **Build Status:** Clean (0 Vite/TypeScript errors, compiled in 1.34s)
- **Hosting:** Live on Vercel at <https://symposium-website-steel.vercel.app>
- **Database & Storage:** Fully verified with check constraints and RLS policies